



Science — Cbse Annual Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	What are skeletal muscle fibres described as? (A) Spindle-shaped, with a single nucleus, lack striations (B) Cylindrical, branched, with a single nucleus, faint striations (C) Long, cylindrical, unbranched, multinucleate, striated (D) Short, branched, multinucleate, lack striations Answer: Correct Option: (C) ► Why? C. Skeletal muscles are made up of bundles of long, cylindrical cells called muscle fibres, which are unbranched, multinucleate and striated. [Ch 3: Tissues in Action]	1
2	Why are weather forecasts less certain further into the future? (A) Scientists do not collect enough data for long-term forecasts (B) Tiny differences in conditions can grow over time and lead to completely different outcomes (C) Measuring instruments are not accurate enough for long-range forecasts (D) Weather patterns repeat in unpredictable cycles Answer: Correct Option: (B) ► Why? B. Weather depends on many changing factors, and very tiny differences in conditions can grow over time and lead to something completely different, making forecasts less certain further ahead. [Ch 1: Exploration: Entering the World of Secondary Science]	1



Q.NO.	QUESTIONS	MARKS
3	Assertion (A): Calcium carbonate deposits around hot springs may have helped in the formation of the first protective membrane. Reason (R): These deposits protected early organic molecules from harmful radiation and extreme conditions. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. The assertion is true as the deposits may have helped form the first protective membrane, and the reason is also true as the deposits protected early organic molecules from harmful radiation. However, the protective function against radiation does not explain how the deposits helped in the formation of the membrane; these are two separate functions described in the text. [Ch 2: Cell: The Building Block of Life]	1
4	Assertion (A): Experiments by Louis Pasteur proved that new life always comes from pre-existing life. Reason (R): For long, people thought living things arose spontaneously from non-living matter. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. The assertion is true because Louis Pasteur's experiments disproved spontaneous generation and proved that new life comes from pre-existing life. The reason is also true but does not explain the assertion, as it merely states the old belief that was disproved by Pasteur. [Ch 11: Reproduction: How Life Continues]	1
5	In a sound wave, what is a compression? (A) A region where density is higher than the average density (B) A region where density is lower than the average density (C) A region where density is equal to the average density (D) A region where no particles exist Answer: Correct Option: (A) ► Why? A. A compression is a region of air with higher density compared to the average density, formed when the piston pushes and compresses nearby air particles. [Ch 10: Sound Waves: Characteristics and Applications]	1



Q.NO.	QUESTIONS	MARKS
6	Why do vegetatively propagated plants tend to be more vulnerable to diseases than sexually reproduced plants? (A) They grow at a slower rate than sexually reproduced plants (B) They produce fewer flowers and fruits than sexually reproduced plants (C) They cannot absorb nutrients from the soil efficiently (D) They have less genetic diversity, so a single disease can affect all the plants Answer: Correct Option: (D) ► Why? D. They have less genetic diversity, so a single disease can affect all the plants. Since vegetatively propagated plants are genetically identical, they share the same susceptibility to a particular disease. <i>[Ch 11: Reproduction: How Life Continues]</i>	1
7	In the hierarchical classification system, which category comes immediately after Class? (A) Order (B) Family (C) Genus (D) Phylum Answer: Correct Option: (A) ► Why? A. The hierarchy is Kingdom → Phylum → Class → Order → Family → Genus → Species, so Order comes immediately after Class. <i>[Ch 12: Patterns in Life: Diversity and Classification]</i>	1
8	When atoms combine, the total energy of the system becomes lower, making the arrangement more stable. What is the force that holds these atoms together called? (A) Nuclear force (B) Gravitational force (C) Magnetic force (D) Chemical bond Answer: Correct Option: (D) ► Why? D. Chemical bond because the text defines the force that holds atoms together, which lowers the total energy of the system, as a chemical bond. <i>[Ch 9: Atomic Foundations of Matter]</i>	1
9	Which organelle is continuous with the outer membrane of the nuclear envelope? (A) Golgi apparatus (B) Endoplasmic reticulum (C) Mitochondria (D) Lysosome Answer: Correct Option: (B)	1



Q.NO.	QUESTIONS	MARKS
	► Why? B. The Endoplasmic Reticulum (ER) is a large organelle that spreads like a network within the cytoplasm and is continuous with the outer membrane of the nuclear envelope. [Ch 2: Cell: The Building Block of Life]	
10	What does an object do when a net force acts on it, according to Newton's second law of motion? (A) Accelerates opposite to the direction of the net force (B) Accelerates in the direction of the net force (C) Moves with constant velocity in the direction of the net force (D) Rotates perpendicular to the net force Answer: Correct Option: (B) ► Why? B. Newton's second law states that when a net force acts on an object, the object accelerates in the direction of the net force. [Ch 6: How Forces Affect Motion]	1
11	Assertion (A): The intermediate points on a straight line position-time graph represent possible values for the position of a vehicle at intermediate times if the vehicle is moving with a constant speed. Reason (R): A position-time graph only represents the positions of an object at the specific instants of time for which data was recorded. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. The assertion is true because if a vehicle moves with constant speed, the continuous line allows intermediate points to accurately represent intermediate positions. The reason is false because a position-time graph can represent positions at intermediate times, not just the specific instants for which data was originally recorded. [Ch 4: Describing Motion Around Us]	1
12	Assertion (A): Two points separated by 0.05 mm cannot be seen as distinct by the unaided human eye from a distance of 25 cm. Reason (R): The limit of resolution of the human eye is 0.1 mm. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A)	1



Q.NO.	QUESTIONS	MARKS
	► Why? Both A and R are true, and R is the correct explanation of A. The limit of resolution of the human eye is 0.1 mm, meaning any two points closer than 0.1 mm cannot be distinguished as separate. Since 0.05 mm is less than 0.1 mm, the reason explains why the points cannot be seen as distinct. [Ch 2: Cell: The Building Block of Life]	
13	An object is said to be at rest if: (A) Its speed is zero. (B) Its position with respect to the reference point does not change with time. (C) Its average velocity is zero. (D) It does not travel any distance. Answer: Correct Option: (B) ► Why? B. An object is at rest if its position with respect to the reference point does not change with time. [Ch 4: Describing Motion Around Us]	1
14	What does the magnifying glass symbolise in the context of scientific exploration? (A) Mathematical precision in calculations (B) Careful observation and noticing patterns (C) Direction in choosing appropriate models (D) Experimental testing of hypotheses Answer: Correct Option: (B) ► Why? B. The magnifying glass symbolises careful observation—noticing patterns and paying attention to what might otherwise be missed. [Ch 1: Exploration: Entering the World of Secondary Science]	1
15	Assertion (A): Mature red blood cells in humans cannot divide or repair themselves. Reason (R): Mature red blood cells lack a nucleus. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? Both A and R are true, and R is the correct explanation of A. The absence of a nucleus in mature red blood cells directly prevents them from repairing or dividing. Therefore, the reason correctly explains the assertion. [Ch 2: Cell: The Building Block of Life]	1



Q.NO.	QUESTIONS	MARKS
16	A girl carries a box while walking on a horizontal surface. No work is done by the upward force she applies on the box because: (A) The force applied is zero (B) The displacement of the box is zero (C) The mass of the box is zero (D) The force is perpendicular to the displacement of the box Answer: Correct Option: (D) ► Why? D. The force is perpendicular to the displacement of the box, because when a force acts in a direction perpendicular to the displacement, the work done by that force is zero. <i>[Ch 7: Work, Energy, and Simple Machines]</i>	1
17	Which scientist reported that all animals are made up of cells? (A) Matthias Schleiden (B) Theodor Schwann (C) Rudolf Virchow (D) Gottlieb Haberlandt Answer: Correct Option: (B) ► Why? B. In 1839, German zoologist Theodor Schwann found that all animals are also made up of cells. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
18	The principle of purification by crystallization is based on: (A) The difference in densities of substances (B) The difference in boiling points of substances (C) The difference in particle sizes of substances (D) The differences in the solubility of a substance at different temperatures Answer: Correct Option: (D) ► Why? D. The principle of purification by crystallization is based on the differences in the solubility of a substance at different temperatures. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
19	In Bohr's model of the atom, what are the fixed circular paths in which electrons revolve called? (A) Parmanus (B) Stationary states or shells (C) Nucleus (D) Dyads Answer: Correct Option: (B)	1



Q.NO.	QUESTIONS	MARKS
	► Why? B. Bohr proposed that electrons revolve in fixed circular paths called stationary states, orbits, or shells. [Ch 8: Journey Inside the Atom]	
20	What happens when excess atmospheric CO₂ is absorbed by the oceans? (A) Sea water becomes more acidic, threatening plankton and coral reefs (B) Sea water becomes more alkaline, promoting coral reef growth (C) Sea water temperature decreases, increasing marine biodiversity (D) Sea water absorbs more oxygen, benefiting fish populations Answer: Correct Option: (A) ► Why? A. Excess atmospheric CO₂ increases ocean absorption, making sea water more acidic, which threatens tiny plankton and coral reefs and disrupts marine ecosystems. [Ch 13: Earth as a System: Energy, Matter, and Life]	1
Section B Attempt 5 out of 7.		
21	How can one scientifically disprove the claim that food becomes harmful during an eclipse? Answer: An eclipse is just a play of shadows. Asking scientific questions reveals no physical, chemical, or biological mechanism supports the claim. Questions about measurable data—like whether temperature changes significantly during an eclipse, or whether food goes bad in a shadow—help disprove it. ► Why? This tests the application of scientific questioning to evaluate a viral claim, as illustrated in the source's eclipse example. [Ch 1: Exploration: Entering the World of Secondary Science]	2
	OR Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg²⁺. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. Answer: Cl⁻, one, two	



Q.NO.	QUESTIONS	MARKS
	► Why? According to the textbook context, a chlorine atom gains one electron to acquire a negative charge, becoming a chloride anion represented as Cl^-. Since a magnesium atom gives two electrons to become Mg^{2+}, and each chlorine atom can only accept one electron, it requires two chlorine atoms to accept the two electrons. Therefore, one ion of magnesium and two ions of chlorine combine to form magnesium chloride, ensuring that the positive and negative charges balance each other to create a neutral overall structure. ► Step-by-step: - Step 1: Identify the ion formed by chlorine. The context states chlorine gains one electron to become a chloride anion, Cl^-. - Step 2: Determine the ratio of ions. Magnesium forms a Mg^{2+} ion by giving two electrons. Since each Cl atom takes only one electron, two Cl^- ions are needed to accept the two electrons from one Mg^{2+} ion. - Step 3: Fill in the blanks with 'Cl^-', 'one', and 'two'. [Ch 9: Atomic Foundations of Matter]	
22	In F.C. Steward's 1958 experiment, what process allows mature phloem cells to form an undifferentiated mass of unspecialised cells before redifferentiating? Answer: The process is dedifferentiation. It allows mature phloem cells to regain the ability to divide, forming an undifferentiated mass of unspecialised cells before they redifferentiate into roots, shoots, and eventually a complete plant. ► Why? Dedifferentiation is the specific term used in the text for cells regaining the ability to divide and becoming unspecialised before redifferentiation occurs. [Ch 3: Tissues in Action]	2
	OR At what minimum difference in boiling points can two miscible liquids be separated by distillation? Answer: Two miscible liquids can be separated by distillation if their boiling points differ by at least about 25 °C. A large difference in boiling points allows one liquid to vaporise before the vapours of the other liquid are formed in significant amount, enabling effective separation. ► Why? The source specifies that distillation works for liquids differing in boiling point by at least about 25 °C, as this ensures one liquid vaporises predominantly before the other. [Ch 5: Exploring Mixtures and their Separation]	
23	Differentiate between Rough Endoplasmic Reticulum (RER) and Smooth Endoplasmic Reticulum (SER) based on their structure and primary function. Answer: RER has ribosomes attached to its surface, making it look rough, and is mainly involved in protein synthesis and secretion. SER lacks ribosomes, appears smooth, and is involved in the synthesis and storage of fats and hormones. ► Why? The question requires distinguishing between two types of Endoplasmic Reticulum by linking their structural differences (presence/absence of ribosomes) to their specific functional roles. [Ch 2: Cell: The Building Block of Life]	2



Q.NO.	QUESTIONS	MARKS
24	An object moves with a constant velocity of 20 m s^{-1} for 6 s. Calculate its displacement using the area under the velocity-time graph. Answer: Displacement = area of rectangle = velocity $\times$ time = $20 \text{ m s}^{-1} \times 6 \text{ s} = 120 \text{ m}$. Since velocity is constant, it equals the average velocity, so the area directly gives displacement. ► Why? The source demonstrates this exact calculation where constant velocity multiplied by time interval gives displacement as the rectangular area under the v-t graph. [Ch 4: Describing Motion Around Us]	2
25	What does the force of friction depend upon? How can a spring balance be used to measure it? Answer: The force of friction depends upon the nature of the surfaces in contact. A spring balance attached to a wooden block is pulled horizontally on different surfaces. The reading when the block just starts moving gives an approximate measure of the force of friction between the surfaces. ► Why? The source describes Activity 6.2 where a spring balance measures friction by noting the reading when the block just begins to move on different surfaces. [Ch 6: How Forces Affect Motion]	2
Section C Attempt 5 out of 7.		
26	40 g of salt is dissolved in 160 g of water. Calculate the mass by mass percentage of the solution formed. Answer: Mass by mass percentage expresses the concentration of a solution by indicating how many grams of solute are present in 100 grams of the total solution. It is also used for heterogeneous mixtures like milk powder. To find the concentration, first calculate the total mass of the solution. The mass of solution equals the mass of solute plus the mass of solvent: $40 \text{ g} + 160 \text{ g} = 200 \text{ g}$. Using the formula, Mass by mass percentage = (Mass of solute / Mass of solution) $\times$ 100. Substituting the values: $(40 \text{ g} / 200 \text{ g}) \times 100 = 20\% \text{ m/m}$. ► Why? The answer requires calculating the total mass of the solution first, and then applying the mass by mass percentage formula provided in the source text. ► Step-by-step: - Step 1: Calculate the total mass of the solution by adding the mass of the solute and the mass of the solvent ($40 \text{ g} + 160 \text{ g} = 200 \text{ g}$). - Step 2: Write down the formula for mass by mass percentage: $(\text{Mass of solute} / \text{Mass of solution}) \times 100$. - Step 3: Substitute the values into the formula $(40 / 200) \times 100$ to get the final answer of $20\% \text{ m/m}$. [Ch 5: Exploring Mixtures and their Separation]	3



Q.NO.	QUESTIONS	MARKS
	OR How does the epidermis protect plants from various environmental factors, and what role does the cuticle play in plants living in dry habitats? Answer: The epidermis forms the outermost layer of the plant body, consisting of a tightly packed, single layer of flat and rectangular cells. It protects all parts of the plant from mechanical injury, water loss, harmful microorganisms, and extreme environmental conditions. In plants living in very dry habitats, the epidermis is covered by a thick waxy layer of cutin called the cuticle. This thick cuticle significantly reduces water loss during transpiration through the stomata. Additionally, the cuticle provides protection against mechanical injury and invasion by parasites, ensuring the plant's survival in harsh, arid environments. ► Why? The epidermis acts as a protective barrier, while the cuticle, a waxy layer, minimizes water loss and protects against injury and parasites, which is crucial for survival in dry habitats. ► Step-by-step: 1. Step 1: Describe the epidermis as a tightly packed outermost layer providing general protection. 2. Step 2: Identify the cuticle as a waxy layer of cutin on the epidermis. 3. Step 3: Explain that in dry habitats, a thick cuticle reduces water loss by transpiration and protects against injury and parasites. [Ch 3: Tissues in Action]	
27	Differentiate between osmosis and diffusion based on their definitions and the requirement of a membrane. Answer: Diffusion is the net movement of particles from an area of higher concentration to an area of lower concentration. This movement occurs due to a concentration gradient and happens even without a membrane. Osmosis, on the other hand, is specifically the diffusion of water across a selectively permeable membrane. It involves the movement of water from an area with more water and less solute to an area with less water and more solute. For example, in plants, water from the soil enters root cells by the process of osmosis. ► Why? Diffusion is the general movement of particles down a concentration gradient without needing a membrane, whereas osmosis is specifically water movement across a selectively permeable membrane. ► Step-by-step: 1. Step 1: Define diffusion and note it does not require a membrane. 2. Step 2: Define osmosis and note it requires a selectively permeable membrane. 3. Step 3: Highlight that osmosis specifically involves the movement of water. [Ch 2: Cell: The Building Block of Life]	3



Q.NO.	QUESTIONS	MARKS
	OR A stack of coins is pushed by a stretched rubber band and released on a wooden table and then on a polished marble floor. On which surface will the coins travel a larger distance before stopping, and why? Answer: The stack of coins will travel a larger distance on the polished marble floor before coming to rest. This happens because the force of friction depends upon the nature of the surfaces in contact. The polished marble floor is smoother than the wooden table top, resulting in a smaller force of friction acting against the motion of the coins. Because the friction is smaller, the velocity of the stack of coins decreases more slowly. Consequently, it travels a greater distance before its velocity reaches zero. ► Why? This requires applying the concept that friction depends on the nature of surfaces, linking smaller friction to a slower decrease in velocity and a larger distance travelled. ► Step-by-step: 1. Step 1: Identify that the force of friction depends on the nature of surfaces in contact. 2. Step 2: Compare the smoothness of the wooden table and the polished marble floor. 3. Step 3: Conclude that the marble floor has less friction, causing velocity to decrease more slowly and the coins to travel a larger distance. [Ch 6: How Forces Affect Motion]	
28	What is a scientific model? Explain with two examples how models simplify real systems by deliberately ignoring certain details. Answer: A scientific model is a simplified way of looking at a real system that focuses only on what is most important for a given question. Building models involves making assumptions and deliberately ignoring certain details to keep things simple. In physics, a moving car may be represented as a single point to study its overall motion, ignoring its shape and size. Similarly, when studying a falling object, air resistance may be neglected to understand the basic effect of gravity. In biology, when studying how the heart pumps blood, many individual cells are ignored so that the organ can be understood as a functioning system. ► Why? Models simplify complex real systems by focusing on key features and deliberately ignoring minor details to make understanding and calculation feasible. ► Step-by-step: 1. Step 1: Define a scientific model as a simplified representation of a real system. 2. Step 2: Give an example from physics (car as a point, ignoring shape/size or neglecting air resistance). 3. Step 3: Give an example from biology (heart as an organ, ignoring individual cells). [Ch 1: Exploration: Entering the World of Secondary Science]	3
29	If a marble moving along the inner boundary of a circular ring is released by lifting the ring, in what direction does it continue to move? What does this indicate about the direction of its velocity at any instant during circular motion?	3



Q.NO.	QUESTIONS	MARKS
	Answer: When the marble is released by lifting the ring, it continues to move in a straight line. This straight-line path is along the tangent to the circle at the point where the marble was released. This demonstrates that in circular motion, the direction of velocity at any instant is along the tangent to the circular path in the direction of motion. Even if the speed is constant, the direction of velocity changes continuously, which means the velocity is changing and the object is accelerating. ► Why? The marble moves tangentially upon release, proving that the instantaneous velocity in circular motion is always directed along the tangent to the circle at that point. ► Step-by-step: - Step 1: Observe that the released marble moves in a straight line. - Step 2: Relate this straight line to the tangent of the circle at the release point. - Step 3: Conclude that velocity direction in circular motion is always along the tangent. [Ch 4: Describing Motion Around Us]	
30	Does the potential energy of an object near the surface of the Earth change if it moves with constant velocity in the horizontal direction? What if the object is gradually raised in the vertical direction? Answer: (i) No, the potential energy of the object does not change. The potential energy of an object near the Earth's surface is given by mgh, which depends on its height. Since moving horizontally does not change the height of the object, its potential energy remains constant. (ii) Yes, the potential energy increases. When the object is gradually raised in the vertical direction, its height above the Earth's surface increases. As potential energy is directly proportional to height (mgh), raising the object to a greater height results in an increase in its potential energy. ► Why? The textbook establishes that the potential energy of an object near the Earth's surface is calculated as mgh, meaning it is entirely dependent on the object's height. For part (i), horizontal motion does not alter the height (h), so the potential energy remains unchanged. For part (ii), gradually raising the object vertically increases its height (h), and since potential energy is directly proportional to height, the potential energy increases accordingly. [Ch 7: Work, Energy, and Simple Machines]	3
Section D Attempt 4 out of 6.		
31	A student said, "The Earth moves around the Sun". In this context, discuss whether an object kept on the Earth can be considered to be at rest.	5



Q.NO.	QUESTIONS	MARKS
	Answer: To determine whether an object kept on the Earth is at rest or in motion, we must first specify a reference point. As we know, the position of an object is described by its distance and direction with respect to a fixed reference point. If we choose the Earth as our reference point, the position of the object kept on it does not change with time. Since its position relative to the Earth remains the same, the object is considered to be at rest. However, the student correctly stated that the Earth moves around the Sun, similar to the motion of planets revolving around the Sun. If we choose the Sun as our reference point, the position of the object changes continuously with time because it is moving along with the Earth in a circular path. Thus, with respect to the Sun, the object is in motion. Therefore, whether an object is at rest or in motion is not absolute; it depends entirely on the chosen reference point. An object can be considered at rest relative to one reference point while being in motion relative to another. ► Why? The textbook establishes that motion and rest are relative concepts determined by the observer's frame of reference. An object is at rest if its position does not change with respect to the reference point, and in motion if it does. Since Earth revolves around the Sun, an object on Earth is stationary relative to Earth but moving relative to the Sun. ► Step-by-step: 1. Step 1: Define the role of a reference point in determining motion or rest. 2. Step 2: Analyze the object's state with Earth as the reference point (at rest). 3. Step 3: Analyze the object's state with the Sun as the reference point (in motion). 4. Step 4: Conclude that rest and motion are relative depending on the chosen reference point. [Ch 4: Describing Motion Around Us]	
	OR Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess.	



Q.NO.	QUESTIONS	MARKS
	Answer: In science, laws, theories, and principles are terms with specific and distinct meanings that help organise our understanding of the world in a systematic way. A law usually describes a regular pattern observed in nature, often expressed using words or mathematical relationships. For example, Newton's laws of motion explain the jerk felt when a bus stops suddenly. Laws describe what happens based on repeated observations. A theory goes a step further than a law. It provides an explanation of why those patterns occur, usually based on evidence gathered over time and available at that time. For example, the atomic theory explains how molecules are formed. In science, a theory does not mean a guess or an untested idea; it is an explanation based on careful testing and critical examination. Principles are broad ideas that help us make sense of a given situation. For example, the principle of conservation of energy can be applied when climbing up the stairs. Principles serve as guiding ideas rather than descriptions or explanations. A key feature that makes science reliable is that these ideas are always open to improvement and often change as new evidence becomes available. No scientific theory is ever final and none is beyond question. This openness to being corrected by nature itself is what allows science to help us understand the world we live in. ► Why? Laws describe observed patterns, theories explain why patterns occur based on evidence, and principles are broad guiding ideas. A scientific theory is a well-tested explanation, not a guess. ► Step-by-step: 1. Step 1: Define a law and give the Newton's laws of motion example. 2. Step 2: Define a theory and give the atomic theory example; clarify it is not a guess. 3. Step 3: Define a principle and give the conservation of energy example. 4. Step 4: Explain how all these ideas remain open to improvement as new evidence emerges. 5. Step 5: Conclude by stating why this openness makes science reliable. [Ch 1: Exploration: Entering the World of Secondary Science]	
32	Describe how a rocket lifts off the ground based on Newton's third law of motion. Also, explain how a spacecraft moving in space where gravitational force is negligible can slow down.	5



Q.NO.	QUESTIONS	MARKS
	Answer: A rocket lifts off based on Newton's third law of motion, which states that forces always occur in equal and opposite pairs acting on different objects. When a rocket's engine operates, it produces gas and expels it in the downward direction. According to Newton's third law, the expelled gas exerts an equal and opposite force on the rocket in the upward direction. This upward force on the rocket becomes larger than the weight of the rocket. As a result, the net force on the rocket acts in the upward direction, causing the rocket to lift off. If a spacecraft is moving in a region of space where the gravitational force is negligible, it can change its velocity by firing its engine. If the engine fires in the direction of its motion, the exhaust gases are expelled forward. These exhaust gases exert a force in the direction opposite to the direction of motion of the spacecraft, thereby slowing it down. This principle was used by the Vikram lander of Chandrayaan-3 to slow down and attain the necessary velocity for a soft landing near the south pole of the Moon. ► Why? <i>The rocket pushes gas downwards, and the gas pushes the rocket upwards. Similarly, firing exhaust in the direction of motion creates an opposing force that slows the spacecraft.</i> ► Step-by-step: 1. Step 1: State how the rocket expels gas downwards, creating an upward reaction force. 2. Step 2: Explain that the upward force exceeds the rocket's weight, creating a net upward force for lift-off. 3. Step 3: Explain how firing an engine in the direction of motion expels gas forward, creating a backward reaction force. 4. Step 4: Conclude that this backward force slows down the spacecraft, as used by the Vikram lander. <i>[Ch 6: How Forces Affect Motion]</i>	
	OR Define totipotency. Describe how F.C. Steward demonstrated this property using carrot phloem cells, and explain the processes of dedifferentiation and redifferentiation that occur during the regeneration of a complete plant.	



Q.NO.	QUESTIONS	MARKS
	Answer: Totipotency is the ability of some mature plant cells to undifferentiate, divide, and redifferentiate to develop into a new plant under specific conditions. This is similar to the ability of a zygote to divide and differentiate into an entire organism. In 1958, F.C. Steward and his team demonstrated totipotency using phloem cells of carrot. They grew these phloem cells in a nutrient medium containing simple sugars and hormones under appropriate conditions. The phloem cells divided to form a mass of cells, which gradually divided and differentiated into a complete plant. The process involves two key stages. First, the mature phloem cells dedifferentiate — they regain the ability to divide — forming an undifferentiated mass of unspecialised cells. Second, when these unspecialised cells are grown in appropriate conditions supplemented with nutrients and growth chemicals, they further divide and redifferentiate to form roots, shoot, and eventually the complete plant. The experimental results also showed that growth conditions matter. Cells grown in a liquid medium with nutrients and air showed a 20% increase in fresh weight, while cells under other conditions showed reduced growth. This confirms that totipotency requires specific conditions of nutrients, air, and growth chemicals to be expressed fully. ► Why? Totipotency is demonstrated by Steward's experiment where carrot phloem cells dedifferentiated into unspecialised cells and then redifferentiated into a complete plant, proving mature plant cells can regenerate under appropriate conditions. ► Step-by-step: 1. Step 1: Define totipotency as the ability of mature plant cells to undifferentiate, divide and redifferentiate into a new plant under specific conditions. 2. Step 2: Describe Steward's experiment — phloem cells of carrot grown in nutrient medium with simple sugars and hormones formed a complete plant. 3. Step 3: Explain dedifferentiation — mature cells regain the ability to divide, forming an undifferentiated mass of unspecialised cells. 4. Step 4: Explain redifferentiation — unspecialised cells, under appropriate conditions with nutrients and growth chemicals, divide and differentiate into roots, shoot, and complete plant. 5. Step 5: Conclude by mentioning that specific conditions (liquid medium, air, nutrients) are essential, as shown by the 20% increase in fresh weight versus reduced growth under other conditions. [Ch 3: Tissues in Action]	
33	Explain the process of osmosis and how it differs from diffusion. Describe the behaviour of cells when placed in isotonic, hypotonic, and hypertonic solutions, using the example of potato pieces in plain water and salt solution.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Osmosis is the diffusion of water across a selectively permeable membrane. Water moves from an area with more water and less solute to an area with less water and more solute until the concentrations in the two areas become equal. Diffusion, in contrast, is the net movement of particles from a higher to a lower concentration and occurs even without a membrane. When a potato piece is placed in plain water (Beaker A), water moves into the potato cells by osmosis because the extracellular medium has less solute than the intracellular medium. This is a hypotonic solution. The potato piece swells and its weight increases. When a potato piece is placed in a 20 per cent salt solution (Beaker B), water moves out of the potato cells because the extracellular medium has more solute than the intracellular medium. This is a hypertonic solution. The potato piece shrinks and its weight decreases. In an isotonic solution, the solute concentration of the extracellular medium equals the solute concentration of the intracellular medium, so there is no net movement of water and the cell maintains its normal state. The key difference between osmosis and diffusion is that osmosis requires a selectively permeable membrane and involves only water movement, while diffusion does not require a membrane and involves the movement of any particles from higher to lower concentration. ► Why? Osmosis is water movement across a selectively permeable membrane driven by concentration differences; cells swell in hypotonic solutions and shrink in hypertonic solutions. ► Step-by-step: 1. Step 1: Define osmosis and diffusion, and state the key difference between them. 2. Step 2: Explain the potato experiment — swelling in plain water (hypotonic) and shrinking in salt solution (hypertonic). 3. Step 3: Define isotonic, hypotonic, and hypertonic solutions with their effects on cells. 4. Step 4: Compare osmosis and diffusion clearly — membrane requirement and type of particles moving. 5. Step 5: Conclude summarising how solution type determines the direction of water movement. [Ch 2: Cell: The Building Block of Life]	
34	Describe the process of paper chromatography for separating the components of a mixture. Explain the principle behind this separation and mention how it can be applied to separate pigments from plant materials.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Paper chromatography is a method of separating the components of a mixture that uses differences in the interactions of the components with the solvent and the paper. The liquid carries the substances up the paper, separating them based on how fast they move. To perform paper chromatography, take a 3 cm wide strip of chromatographic paper or filter paper. Draw a straight, horizontal line with a pencil 2 cm from the bottom. Mark a spot with a black sketch pen at the centre of this line. Place the paper strip vertically into a container, like a gas jar or beaker, with a thin layer of water at the bottom. Ensure the water level is below the ink spot. As the water rises through the paper, the ink starts to separate into different colour spots. This separation occurs because different coloured components of the ink move up the paper at different speeds based on their interactions with the solvent and the paper. This technique can be applied to separate various pigments from natural sources. For example, you can separate different pigments present in the green extract of spinach leaves or coloured pigments from flower petals. While water is commonly used as a solvent, it does not work in every case. Sometimes, a different solvent is required, such as a 2% m/v salt solution, alcohol, or a mixture of solvents, depending on the substances being separated. Choosing the correct solvent ensures that the components separate effectively on the chromatographic paper, allowing for clear identification. ► Why? Paper chromatography separates mixtures based on the different rates at which components travel up a paper strip via a solvent, which allows for the isolation of various pigments in substances like ink or spinach leaves. ► Step-by-step: 1. Step 1: Define paper chromatography and its underlying principle. 2. Step 2: Describe the step-by-step procedure for setting up a paper chromatography experiment. 3. Step 3: Explain what happens as the water rises through the paper and why the separation occurs. 4. Step 4: Give examples of natural substances that can be separated using this technique. 5. Step 5: Discuss the role of different solvents when water is not effective. [Ch 5: Exploring Mixtures and their Separation]	
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Cell Membrane and Osmosis A student set up an experiment with two potato pieces of roughly equal size and equal initial weight. One piece was placed in Beaker A containing plain water, and the other in Beaker B containing a 20 per cent salt solution. After an hour, the piece in Beaker A swelled and its weight increased, while the piece in Beaker B shrank and its weight decreased. The student learned that the cell membrane is extremely thin, about 7 to 10 nm thick, and is described by the fluid-mosaic model. In this model, the membrane has a lipid bilayer with proteins embedded in them acting as gatekeepers, and the molecules can move sideways, flip, and rotate within the membrane. (i) Why did the potato piece in Beaker A swell when placed in plain water? (ii) What is the role of proteins embedded in the lipid bilayer of the cell membrane? (iii) Explain the process that caused the potato piece in Beaker B to shrink and lose weight when placed in a 20 per cent salt solution. (iv) How does the fluid-mosaic model explain the 'fluid' nature of the cell membrane? Answer: (i) Water moved from an area with more water and less solute (plain water) to an area with less water and more solute (inside the cell) through osmosis. (ii) They act like gatekeepers in helping substances pass through the membrane. (iii) The 20 per cent salt solution has a higher solute concentration than the intracellular medium. Water moves out of the potato cells to the concentrated solution across the selectively permeable membrane by osmosis, causing the cells to shrink and lose weight. (iv) The molecules in the lipid bilayer can move sideways, flip, and rotate within the membrane, making it fluid. ► Why? i) Water moved from an area with more water and less solute (plain water) to an area with less water and more solute (inside the cell) through osmosis. ii) They act like gatekeepers in helping substances pass through the membrane. iii) The 20 per cent salt solution has a higher solute concentration than the intracellular medium. Water moves out of the potato cells to the concentrated solution across the selectively permeable membrane by osmosis, causing the cells to shrink and lose weight. iv) The molecules in the lipid bilayer can move sideways, flip, and rotate within the membrane, making it fluid. [Ch 2: Cell: The Building Block of Life]	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: Kinematic Equations for Vehicles on a Straight Road A motorbike is moving on a straight road with an initial velocity of 28 m/s. The rider applies brakes and the motorbike decelerates uniformly, coming to rest after travelling a distance of 98 m. On a parallel track, a car starts from rest and accelerates uniformly, reaching a final velocity of 24 m/s in 6 seconds. A physics student watching both vehicles decides to compare their accelerations and the distances they cover using the kinematic equations for motion in a straight line with constant acceleration. (i) What is the average acceleration of the car during the 6 seconds? (ii) Using the equation $v^2 = u^2 + 2as$, calculate the acceleration of the motorbike. (iii) Calculate the distance travelled by the car in 6 seconds and the time taken by the motorbike to come to a stop. (iv) Compare the magnitudes of acceleration of the motorbike and the car. Which one is greater? Answer: (i) 4 m/s² ($a = (v - u)/t = (24 - 0)/6 = 4 \text{ m/s}^2$) (ii) -4 m/s² ($0^2 = 28^2 + 2a(98) \rightarrow 0 = 784 + 196a \rightarrow a = -4 \text{ m/s}^2$. The negative sign indicates deceleration; magnitude is 4 m/s².) (iii) For the car: $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(4)(6^2) = 72 \text{ m}$. For the motorbike: $v = u + at \rightarrow 0 = 28 + (-4)t \rightarrow t = 7 \text{ s}$. (iv) Both have the same magnitude of acceleration (4 m/s²). Neither is greater. ► Why? i) 4 m/s² ($a = (v - u)/t = (24 - 0)/6 = 4 \text{ m/s}^2$) ii) -4 m/s² ($0^2 = 28^2 + 2a(98) \rightarrow 0 = 784 + 196a \rightarrow a = -4 \text{ m/s}^2$. The negative sign indicates deceleration; magnitude is 4 m/s².) iii) For the car: $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(4)(6^2) = 72 \text{ m}$. For the motorbike: $v = u + at \rightarrow 0 = 28 + (-4)t \rightarrow t = 7 \text{ s}$. iv) Both have the same magnitude of acceleration (4 m/s²). Neither is greater. [Ch 4: Describing Motion Around Us]	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Building Scientific Models A student wants to model how long it takes for a ball thrown upward to return to the ground. She considers several factors: the mass of the ball, its initial speed and direction, air resistance, the colour of the ball, and the spin of the ball. She recalls that physicist Meghnad Saha simplified the study of stars by ignoring complex processes and focusing only on temperature, pressure, and how atoms formed ions, which helped explain the connection between star colour and temperature. She decides to apply a similar simplification approach. She also notes that science uses precise symbols like m for mass, v for velocity, and F for force, and that equations are compact statements about relationships between quantities. (i) What three factors did Meghnad Saha focus on to explain the colour of stars? (ii) Why should the student ignore the colour of the ball when modeling the time it takes to return to the ground? (iii) The student ignores air resistance and spin in her simple model. Later, she wants a more accurate prediction. What should she do, based on the approach described for building complex models? (iv) The student writes an equation relating distance, time, and velocity to describe the ball's motion. Explain why this equation is more than just a calculation tool. Answer: (i) Temperature, pressure, and how atoms formed ions. (ii) The colour of the ball does not affect the motion of the ball, just as the colour of a cricket ball makes no difference to whether it crosses the boundary. Only factors that affect the outcome should be included. (iii) She should add extra details such as air resistance and spin for greater accuracy. As we build more and more complex models, we add extra details beyond the simple model to improve the accuracy of our predictions. (iv) An equation is a compact statement about how certain things are related. Mathematics in science is a language that helps us think more clearly about the world, not just a tool for finding numerical answers. ► Why? i) Temperature, pressure, and how atoms formed ions. ii) The colour of the ball does not affect the motion of the ball, just as the colour of a cricket ball makes no difference to whether it crosses the boundary. Only factors that affect the outcome should be included. iii) She should add extra details such as air resistance and spin for greater accuracy. As we build more and more complex models, we add extra details beyond the simple model to improve the accuracy of our predictions. iv) An equation is a compact statement about how certain things are related. Mathematics in science is a language that helps us think more clearly about the world, not just a tool for finding numerical answers. [Ch 1: Exploration: Entering the World of Secondary Science]	5



Q.NO.	QUESTIONS	MARKS
37	Case Study: Connective Tissues and Blood Components Rohan, a 15-year-old student, was playing football when he fell and scraped his knee. He noticed red blood oozing from the cut, which eventually formed a clot. Upon examining his elbow, he felt a hard, rigid structure. The doctor explained that both blood and bones are connective tissues, but they differ in their matrix — blood has a watery, soft, jelly-like matrix while bone has a hard, rigid matrix containing calcium and phosphorus compounds. The doctor also noted that at the injured site, white blood cells collected, causing inflammation and redness. (i) What gives blood its red colour? (ii) Why do blood and bone differ in consistency despite both being connective tissues? (iii) If platelets were absent in Rohan's blood, what problem would occur at the site of his cut? What role do white blood cells play at the injury site? (iv) Both blood and bone are classified as connective tissues. How does their differing matrix composition relate to their specific functions in the body? Answer: (i) The red colour of blood is due to haemoglobin, an iron-rich protein present in the Red Blood Cells (RBCs). (ii) Blood and bone differ in consistency because of their matrix. Blood has a watery, soft and jelly-like matrix making it fluid, while bone has a hard, solid and rigid matrix containing calcium and phosphorus compounds, making it hard. (iii) Without platelets, Rohan's blood would not clot at the injury site, leading to continuous bleeding. White blood cells collect at the infected area, causing inflammation (redness and swelling) and possible pus formation. (iv) Blood's watery, jelly-like matrix allows it to remain fluid and transport nutrients, gases, and hormones throughout the body. Bone's hard, rigid matrix containing calcium and phosphorus compounds provides strength, support and protection to the body. ► Why? i) The red colour of blood is due to haemoglobin, an iron-rich protein present in the Red Blood Cells (RBCs). ii) Blood and bone differ in consistency because of their matrix. Blood has a watery, soft and jelly-like matrix making it fluid, while bone has a hard, solid and rigid matrix containing calcium and phosphorus compounds, making it hard. iii) Without platelets, Rohan's blood would not clot at the injury site, leading to continuous bleeding. White blood cells collect at the infected area, causing inflammation (redness and swelling) and possible pus formation. iv) Blood's watery, jelly-like matrix allows it to remain fluid and transport nutrients, gases, and hormones throughout the body. Bone's hard, rigid matrix containing calcium and phosphorus compounds provides strength, support and protection to the body. [Ch 3: Tissues in Action]	5

